OOS Number: OOS- 262064

Phase I

Section A – Test Details					
Initiator:	Gabrielle Surber (Written by: Qiyue Chen)	**Test Date:**	25-Aug-2026	**Date Initiated:**	04-Sep-2026
Date of Incident:	04-Sep-2026	**Test Name:**	USP <71> / EP 2.6.1 Sterility Test	**Sample ID:**	ETX-260804-0335
Name of Analyst who Performed the Test:	Prepping Analyst: Elysse Nioupin (EN) Processing Analyst: Gabrielle Surber (GS) Reading Analyst: Elysse Nioupin (EN)	**Sample / Active Name**	Tirzepatide 10mg/mL	**Lot #:**	080326-TIR-10
		Dosage Form	Liquid		
Description of Incident:	On 04 Sep 2026, sample ETX-260804-0335 was found positive for viable microorganisms after USP <71> sterility testing.	☑ Client Care notified of OOS ☑ Sterility and Endotoxin test related OOS; Client must be notified within 24 hours of OOS			
	SOP / Test Method #: MICRO-SOP-5	Effective Date: 24Jul2026	SOP / Test Method Rev: 19	Limits/Specification: Pass/Fail	
Manager notified	Manager/designee name: Kathan Parikh	Date: 04-Sep-2026	Quality Assurance Department notified:	Yes ☑ No ☐	

Section B – Investigation and Root Cause Analysis			
Personnel Discrepancy		**Comments**	
Analyst interviewed?	☑ Yes ☐ No ☐ N/A	Yes, analysts Elysse Nioupin and Gabrielle Surber were interviewed comprehensively.	
Correct samples analyzed?	☑ Yes ☐ No ☐ N/A	Yes, sample ID: ETX-260804-0335	
Correct SOP / Test Method used/followed?	☑ Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-5	
Correct sample collection/technique?	☑ Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-5	
Are the Test Result Records Attached & Calculations Verified and Correct?	☑ Yes ☐ No ☐ N/A	Yes, Information is available in EagleTrax under ETX-260804-0335	
Is the analyst qualified to perform the test?	☑ Yes ☐ No ☐ N/A	Yes, the analysts are trained and qualified by quality to perform the test	
Is formula worksheet provided and verified?	☐ Yes ☐ No ☑ N/A	Not Applicable	
Does the processing method/method set conform to the required specification?	☐ Yes ☐ No ☑ N/A	Not Applicable	
Was the sample stored appropriately?	☑ Yes ☐ No ☐ N/A	Yes, the sample was stored as per client's instructions. Information is available in EagleTrax Sample Location History under ETX-260804-0335	
Material Discrepancy		**Lot Number**	**Expiry Date**
Is the integrity of the sample(s) questionable?	☐ Yes ☑ No ☐ N/A		
Was the sample transported appropriately?	☑ Yes ☐ No ☐ N/A		
Are the other test results from this run under investigation?	☐ Yes ☑ No ☐ N/A		

OOS Number: OOS- 262064

Appropriate/correct glassware used?	☐ Yes ☐ No ☑ N/A		
Are reagents stored in proper condition and not expired?	☑ Yes ☐ No ☐ N/A	Process Consumables: Information is available in EagleTrax Sample Consumables section Environmental Plates: TSA and Surface Plate: see attached environmental logs Monthly Cleaning: H2O2 strips and IPA: See attached monthly cleaning logs	Process Consumables: Information is available in EagleTrax Sample Consumables section Environmental Plates: TSA and Surface Plate: see attached environmental logs Monthly Cleaning: H2O2 strips and IPA: See attached monthly cleaning logs
Standard Used	Not applicable	Not applicable	Not applicable
Solvent Used	Not applicable	Not applicable	Not applicable
Is the standard the source of the root cause?	☐ Yes ☐ No ☑ N/A		
Has the analyst been instructed to keep the original test solution for possible retesting?	☐ Yes ☐ No ☑ N/A		
Equipment Discrepancy		**Equipment ID**	**Calibration Due Date**
Was the instrument set up correctly?	☑ Yes ☐ No ☐ N/A	ISO 5 BSC E001316	Jun 2027
Is the clean room facility certified?	☑ Yes ☐ No ☐ N/A	CR 114 (E001736)	Dec 2026
System Performance Check (if applicable)?	☐ Yes ☐ No ☑ N/A	Not applicable	Not applicable
System Suitability RSD < 2% (PDA)?	☐ Yes ☐ No ☑ N/A	Not applicable	Not applicable
System Suitability RSD < 10% (QDA, CAD)?	☐ Yes ☐ No ☑ N/A	Not applicable	Not applicable
Do the environmental monitoring results conform to required standards?	☑ Yes ☐ No ☐ N/A	See Phase I Summary	See Phase I Summary
Do the controls (i.e., positive, negative, etc.) conform to the required standards?	☑ Yes ☐ No ☐ N/A	See Phase I Summary	See Phase I Summary
Is equipment/instrument functional and calibrated/qualified/certified/PM?	☑ Yes ☐ No ☐ N/A	Incubator E001356 (Monitored by Sensor E001450) Incubator E001357 (Monitored by Sensor E001449) Incubator E001034 (Monitored by Sensor E001501) Incubator E001031 (Monitored by Sensor E001505) ISO 5 BSC E001316	Jan 2027 Feb 2027 Jan 2027 Feb 2027 Aug 2027 Feb 2027 Aug 2027 Feb 2027 Jun 2027

OOS Number: OOS- 262064

Are chromatogram/spectra/raw data correct?	☐ Yes ☐ No ☑ N/A	Not applicable	Not applicable

Section C – Evaluation of Laboratory Testing/Hypothesis Testing (if applicable)

- ☑ N/A
- ☐ Re-injection of sample vial
- ☐ Re-injection of original standard vial
- ☐ Re-vialing of original working standard solution
- ☐ Re-vialing of original working sample solution
- ☐ Re-dilution of a working sample solution from the original stock
- ☐ Re-dilution of a working standard solution from the original stock standard
- ☐ Other: N/A QYC 06Oct26

Section D – Phase I Summary

Phase I Summary:

This investigation was performed as per MICRO-SOP-53 Sterility Test Out-of-Specification (OOS) Investigation Procedure. All analysts involved in the prepping and reading (Elysse Nioupin), and processing (Gabrielle Surber) were interviewed comprehensively. Their answers are recorded throughout this document.

Upon arrival, the sample was stored in accordance with the Client's instructions. Analysts - Elysse Nioupin and Gabrielle Surber - verified the integrity of the sample throughout both the preparation and processing stages. No leaks or turbidity were observed at any point, verifying the integrity of the sample.

All reagents and supplies mentioned in the material section above were stored according to the suppliers' recommendations, and their integrity was visually verified before utilization. Moreover, all reagents and supplies had valid expiration dates. The functionality of all equipment was confirmed by reviewing data generated by our comprehensive in-house continuous monitoring system.

The cleanroom suite (CR 114) used for processing procedures comprises three interconnected sections: the innermost ISO 7 cleanroom (114B), which connects to the middle ISO 7 buffer room (114A), and then to the outermost ISO 8 anteroom (114). A positive air pressure system is maintained throughout the suite to ensure controlled, unidirectional airflow from 114B through 114A and into 114.

The ISO 5 BSC E001316, located in the innermost ISO 7 room, (114B), was used for sample processing steps. It was thoroughly cleaned and disinfected prior to each procedure in accordance with MICRO SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). Additionally, BSC E001316 was certified and approved by both the Engineering and Quality Assurance teams. Sample processing was conducted in the ISO 5 BSC E001316 in the innermost ISO 7 room, (114B) by Gabrielle Surber on 25 Aug 2026 as per MICRO-SOP-5 (USP <71> / EP 2.6.1 Sterility Test).

On 21 Aug 2026, the sample vials for ETX-260804-0335 were received from the Sample Submissions team and brought into the Sterile Microbiology lab. Upon arrival, each sample vial was sprayed with an acidified bleach disinfectant, placed into pre-disinfected bins, and allowed a 10-minute contact time.

The secondary disinfection happened in the ISO 8 anteroom (114), where the vials were again treated with acidified bleach and provided a 10-minute contact time before processing. Subsequently, the vials were moved into innermost ISO 7 cleanroom (114B). Inside this cleanroom, the processing analyst, Gabrielle Surber, performed a final disinfection step, allowing an additional 10-minute contact time. Once fully disinfected, the vials were transferred into the ISO 5 BSC E001316.

Inside the BSC, each container was aseptically opened, ensuring minimal exposure to the environment. The contents of 10 vials were dispensed directly from the container into sterile canisters placed on a vacuum manifold for membrane filtration under method suitability ETX-260811-0813. The sample was processed and tested as per MICRO-SOP-5 (USP <71> / EP 2.6.1 Sterility Test). Media bottles were transferred into incubators E001356 and E001357 to begin incubation.

On 04 Sep 2026 (Day 10 of incubation), microbial growth was observed in one TSB (Tryptic Soy Broth) media bottle by reading analyst, Elysse Nioupin, and confirmed by Microbiology Lab Supervisor, Robin Seymour. The positive TSB bottle for ETX-260804-0335 was submitted for Differential Staining and Microbial Identification under ETX-260904-0331. Definitive identification identified the isolate as Bacillus megaterium, Gram (+) rods (see Table1)

The culture media used were within expiry. Negative control media bottles were processed and incubated under identical conditions and showed no microbial growth

OOS Number: OOS- 262064

Phase I Summary (continued):

ISO 5 BSC Environmental Monitoring Results Evaluation:
After reviewing the Environmental Monitoring (EM) results for the relevant testing period, no microbial growth was detected on personnel monitoring plates, settling plates, or ISO 5 BSC surface sampling plates on the date of testing, the preceding sampling date, or the subsequent sampling date.

Cleanroom Environmental Monitoring Results Evaluation:
No microbial growth was recovered from the weekly active surface sampling plate collected on 25 Aug 2026. However, 4 CFUs (ETX-260901-0112) were recovered from the weekly active air sampling of ISO 8 cleanroom 114. The organisms were identified to be of Gram-positive cocci and hyphae. It is important to note that the environmental isolates were not identical to microorganism identified in test sample (Bacillus megaterium, Gram (+) rods). All sample processing activities were performed within the validated ISO 5 BSC E001316 located in the innermost ISO 7 cleanroom (114B) separated by ISO 7 buffer room (114A) and ISO 8 anteroom (114). Furthermore, the test samples are transferred in disinfected and lidded containers through the layered cleanroom suite. Therefore, as the recovered organisms were different and originated outside the sample processing environment, the cleanroom environmental isolates were not considered related to the test sample contamination.

The absence of contamination on analyst glove plates and work-surface environmental monitoring plates collected on the date of testing indicates that there was no viable transfer of contaminants to the ISO 5 biological safety cabinet (BSC). Additionally, the lack of contamination in other samples processed during the same testing session further demonstrates that the testing environment was operating in optimal condition.

Therefore, considering that no viable organisms were recovered from the ISO 5 environment, personnel monitoring results were acceptable on the date of testing, and the environmental isolates were not recovered from locations or surfaces directly involved in sample preparation or testing, the environmental isolate is not considered a credible source of the contamination detected in the test sample.

The analysts confirmed full compliance with cleaning procedures as outlined in MICRO SOP-9 (Cleaning and Disinfecting Procedure for Microbiology) and MICRO-SOP-5 (USP <71> / EP 2.6.1 Sterility Test).

Monthly cleaning and disinfection of the outermost ISO 8 Anteroom, the middle ISO 7 Buffer room, the innermost ISO 7 cleanroom, and its containing ISO 5 BSCs for Cleanroom suite CR114 was performed on 16 Aug 2026, as per MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). It was documented that all H2O2 indicators passed. This confirms the efficient monthly cleaning of all three parts of Cleanroom suite CR114.

Analyzing a 6-month sample history for Blue Ocean Rx, LLC (E75110), this specific analyte "Tirzepatide 10mg/mL" has had no prior failures using USP <71> / EP 2.6.1 Sterility Test during this period.

To assess the potential for sample-to-sample contamination contributing to the positive results, a comprehensive review was conducted of all samples processed on the same day (25 Aug 2026). Sample ETX-260804-0335 was the 11th sample processed in the batch. All other samples processed by the analyst and other analysts that day tested negative. These findings suggest that cross-contamination between samples is highly unlikely.

Based on the observations outlined above, it is unlikely that the failing results were due to reagents, supplies, the cleanroom environment, the process, or analyst involvement. Consequently, the possibility of laboratory error contributing to this failure is minimal and the original result is deemed to be valid.

OOS Number: OOS- 262064

Phase I Summary (continued):

N/A QYC 06Oct26

OOS Number: OOS- 262064

Has root cause been identified as a possible laboratory error?
Note: Root cause must be conclusive, verifiable, and well-documented.

☐ Yes
- ☐ If retesting is ***required***, complete Phase II of the OOS (initiate Form 3.100.019.F02).

☑ No, the OOS result is valid.

Reviewer comment(s):

Prepared by (*Print*)**:** Qiyue Chen	Signature/Date:
Lab Manager/Designee (*Print*)**:** Robin Seymour	Signature/Date:

Section E – Quality Disposition

Can investigation be closed?

☐ Yes
- ☐ Original result is valid.

☐ No, initiate Form 3.100.019.F02.

Approved by QA Representative (*Print*)**:**	Signature/Date:

Table 1: Information for ETX-260804-0335 under investigation

Processing Analyst	**Reading Analyst**	Sample ID	Related Microbial ID	Media with microbial growth	Microbial ID
Gabrielle Surber	Elysse Nioupin	ETX-260804-0335	ETX-260904-0331	1 x 100mL TSB bottle	*Bacillus megaterium* (Gram (+) rods)

Table 2: Environmental Monitoring (EM) from Processing Performed on 25Aug26

Environmental Monitoring (EM) Sampling Site	**Frequency**	**Date (DDMMMYY)**	**Analyst (Initials)**	**Day /Week(s)**	**Observation**	**Environmental Monitoring Plate ETX ID**	**Microbial ID**	**Notes**
Personnel EM Bracketing								
Personnel (Left Touch and Right Touch)	Daily	21Aug26	GS	Date Before Testing	No growth	N/A	N/A	None
Personnel (Left Touch and Right Touch)	Daily	25Aug26	GS	Date of Testing	No growth	N/A	N/A	None
Personnel (Left Touch and Right Touch)	Daily	26Aug26	GS	Date After Testing	No growth	N/A	N/A	None
Biological Safety Cabinet EM for E001316								
Surface Sampling of ISO 5 (4 locations)	Daily	24Aug26	ES	Date Before Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	25Aug26	GS	Date of Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	26Aug26	GS	Date After Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	24Aug26	ES	Date Before Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	25Aug26	GS	Date of Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	26Aug26	GS	Date After Testing	No growth	N/A	N/A	None
Weekly Active Air Sampling of Anteroom and Cleanroom (E001736, CR 114)								
Active Air Sampling of Cleanrooms	Weekly	25Aug26	SMO	Week of Testing	4 CFU (ISO 8 114)	ETX-260901-0112	3 Gram (+) cocci, 1 Hyphae	None
Surface Sampling of Anteroom and Cleanroom (E001736, CR 114)								
Surface Sampling of Cleanrooms	Weekly	25Aug26	SMO	Week of Testing	No growth	N/A	N/A	None